

SmartGym Adaptive Training System

Start Here

A quick map for finding every SmartGym subsystem and guide.

Version 1.0 Prototype

ESP32-S3 firmware | Firebase RTDB | React dashboard

Generated documentation package

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Language: English | [Español](START_HERE.es.md)

This page is the fastest way to find the right SmartGym information without digging through folders.

What SmartGym Is

SmartGym is an ESP32-S3 prototype that instruments a selectorized gym machine.

It reads machine motion, normalizes range of motion, detects reps and sets, records user sessions, stores calibration data, and syncs finished workouts to Firebase for dashboard analysis.

First Read These

| Need | Open |

| --- | --- |

| Understand the project in 5 minutes | [Project README](../README.md) |

| Use the device during a demo | [User Manual](USER_MANUAL.md) |

| Understand ROM, calibration, equations, and errors | [Measurement and Calibration Guide](MEASUREMENT_AND_CALIBRATION_GUIDE.md) |

| Understand firmware modules and data flow | [System Architecture](SYSTEM_ARCHITECTURE.md) |

| Build or modify firmware | [Developer Guide](DEVELOPER_GUIDE.md) |

| Understand Firebase paths and upload behavior | [Firebase Guide](FIREBASE_GUIDE.md) |

| Understand dashboard seed/test data | [Firebase Dashboard Seed Guide](FIREBASE_DASHBOARD_SEED_GUIDE.md) |

| Fix common hardware, sync, or UI problems | [Troubleshooting Guide](TROUBLESHOOTING.md) |

| Find CAD and printable hardware files | [Hardware CAD README](../hardware/cad/README.md) |

| Run the React dashboard | [Dashboard README](../dashboard/README.md) |

| Test the separate VL53L0X prototype | [Weight Detection README](../weight_detection/README.md) |

If You Are Building the Full Project

Follow this order:

1. Read the [Project README](../README.md).
2. Build the firmware using the commands in the README.
3. Read the [Measurement and Calibration Guide](MEASUREMENT_AND_CALIBRATION_GUIDE.md).
4. Wire and mount hardware using [Hardware Notes](../hardware/README.md) and [CAD files](../hardware/cad/README.md).
5. Configure Firebase using the [Firebase Guide](FIREBASE_GUIDE.md).
6. Use the [Firebase Dashboard Seed Guide](FIREBASE_DASHBOARD_SEED_GUIDE.md) if you need demo data.
7. Run the dashboard using [Dashboard README](../dashboard/README.md).
8. Use the [Troubleshooting Guide](TROUBLESHOOTING.md) during hardware tests.

How the Main Data Flow Works

Motion sensor -> ESP32-S3 -> ROM normalization -> rep detection
 RFID -> user profile -> calibration -> recommendation
 Session recorder -> NVS queue -> Firebase RTDB -> React dashboard

Core Concepts

| Concept | Short Explanation | Full Details |

| --- | --- | --- |

| ROM | The user's calibrated movement range, normalized from bottom to top. |
[Measurement and Calibration Guide](MEASUREMENT_AND_CALIBRATION_GUIDE.md) |

| Pin load | The physical machine weight-pin position. It belongs to the machine, not the user. | [Measurement and Calibration Guide](MEASUREMENT_AND_CALIBRATION_GUIDE.md) |

| Recommended load | User-specific advice from calibration or session history. It does not move the pin. | [Measurement and Calibration Guide](MEASUREMENT_AND_CALIBRATION_GUIDE.md) |

| Rep detection | A ROM state machine that detects bottom, ascent, top, descent, and completion. | [Developer Guide](DEVELOPER_GUIDE.md) |

| Firebase queue | Local NVS-backed retry queue for safe uploads. | [Firebase Guide](FIREBASE_GUIDE.md) |

| Dashboard | React app that reads Firebase workout history. | [Dashboard README](../dashboard/README.md) |

Where Files Live

| Area | Folder |

| --- | --- |

| ESP32-S3 firmware | 'src/', 'lib/', 'boards/', 'platformio.ini' |

| Documentation | 'docs/' |

| Dashboard | 'dashboard/' |

| CAD and STL files | 'hardware/cad/' |

| Hardware notes | 'hardware/' |

| VL53L0X prototype sketch | 'weight_detection/' |

| Utility scripts | 'scripts/', 'tools/' |

| Firebase/sample data | 'sample_data/' |

What To Check Before a Demo

1. Firmware builds and uploads.
2. Wi-Fi connects.
3. RFID scans a known user.
4. User sync finishes before START is enabled.
5. ROM calibration is loaded or a safe fallback is shown.
6. Pin load and recommendation are displayed separately.
7. Session starts, reps count, and summary appears.
8. Firebase upload reaches complete.
9. Dashboard shows the session.

Most Common Confusions

| Confusion | Correct Mental Model |

| --- | --- |

| Recommendation changed, so pin load changed. | No. Recommendation is advice; pin load is the physical machine state. |

| A movement always counts as a rep. | No. A rep must satisfy ROM, timing, and phase-order checks. |

| Firebase upload happens immediately in one request. | No. It is queued locally and uploaded in phases. |

| The dashboard owns the data model. | No. Firmware writes the data; dashboard reads it. |

| The VL53L0X prototype automatically controls the main firmware. | No. It is a separate prototype unless integrated intentionally. |

Security Reminder

Do not commit real Wi-Fi passwords, Firebase secrets, service accounts, tokens, or private `.env` files. Use example files and local environment variables.